Improving the Efficiency of 4-Dimensional SQIsign

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Abstract. SQIsign has the smallest combined public key and signature of any known post-quantum signature. Its dimension-2 signature carries an auxiliary curve whose only purpose is to make the response computable in dimension 2; SQIsignHD's dimension-4 response needs no such curve. We take SQIsignHD's construction to the round-3 SQIsign primes, adopted in September 2026 after Wesolowski's attack, derive its parameters for them, and implement it in our Rust port of the SQIsign reference code. At NIST level I the dimension-4 signature takes 142 bytes, or 139 with the bit-packed layout we give, against 200 for the specification's format; with the 84-byte public key, key and signature together take 226 bytes, or 223, the smallest level-I combination of any post-quantum signature we are aware of at parameters that stand after that attack. The price is verification: measured in one session, the dimension-4 verifier costs 6.4 times the dimension-2 verifier of the same code (85.4 against 13.3 megacycles), or 4.6 times with two kernel points pushed instead of four (60.6 megacycles). We also give a verifier that pushes two kernel points instead of four along the chain, and prove that the algorithm it implements computes the same isogenies as the four-point one.

Keywords: SQIsign · SQIsignHD · isogeny-based cryptography · post-quantum signatures · signature size

1 Introduction

SQIsign is a signature scheme whose public key is an elliptic curve and whose signature describes a path of curves. Its first version had the smallest signatures of any post-quantum proposal and verified fast, but signed in seconds, with a signer built on the KLPT algorithm for finding paths in the quaternion algebra of the curve [**?**].

In 2022 three groups broke the SIDH key exchange by recovering an isogeny between elliptic curves from its action on a few points, embedding it, by Kani's lemma [**?**], in an isogeny between higher-dimensional abelian varieties [**?**, **?**, **?**]. The attack became a tool for SQIsign: the signer no longer needs a path of smooth degree, since any isogeny of known degree can be handed to the verifier through its action on points and rebuilt in higher dimension.

SQIsignHD did this in dimension 4 [**?**]. Its signer is fast and its signature small, and the cost moves to the verifier, which computes a chain of dimension-4 isogenies; its authors released a C signer and a SageMath verifier [**?**]. SQIsign2D-West brought the response back to dimension 2 [**?**]: the signer computes one more isogeny, the auxiliary one, so that the verifier's chain runs on a product of two elliptic curves. Verification became fast again at the cost of sending the auxiliary curve, and this design has been SQIsign's submission to NIST since its second round [**?**, **?**].

In 2026 Wesolowski gave an algorithm for the supersingular isogeny problem in time and memory $p^{1/3+o(1)}$ [**?**]. The round-3 submission of September 2026 answered by replacing the round-2 primes with larger ones, $3 \cdot 2^{324} - 1$, $27 \cdot 2^{500} - 1$ and $17 \cdot 2^{664} - 1$ at levels I, III and V [**?**].

Contributions This paper takes SQIsignHD's dimension-4 response to those primes. We start from sqisign-rs, our Rust implementation of SQISIGN, whose round-3 dimension-2 signer and verifier reproduce the reference's known-answer vectors byte for byte [**?**]. We

- derive the dimension-4 parameters for the round-3 primes, and build the signer and verifier at level I behind a feature flag (Section 4);
- check our signatures against the SQIsignHD authors' own verifier, re-parameterized to our prime;
- make explicit which matrix entry is dropped and how it is recovered, and give a bit-packed encoding that saves three more bytes (Section 4);
- measure everything in one session, and compare our verifier's cost with MIKE's dimension-4 chains (Sections 5 and 7);
- implement a verifier that uses an automorphism of order 4 to push two kernel points instead of four. It agrees with the four-point verifier on every signature we tested, and we prove that the algorithm it implements computes the same isogenies as the four-point one for every input (Appendix A); that our implementation conforms to that algorithm is checked only in part (Remark 3).

Organization Section 2 defines the terminology and Section 3 reviews related work. Section 4 describes the compact format and derives its parameters, Section 5 reports the measurements, and Section 6 explains why dimension 4 does not make SQISIGN strongly unforgeable. Section 7 concludes, with the two-point verifier, the limitations of this work and the steps that remain. Appendix A proves the symmetry the two-point verifier relies on.

2 Terminology

auxiliary isogeny
: In dimension 2, a second isogeny the signer computes next to the response so that together they define a $(2^f, 2^f)$-isogeny the verifier can compute. Its target, the auxiliary curve, is sent in the signature. Dimension 4 has none.

challenge
: The isogeny from the public-key curve selected by hashing the message and the commitment; the signer must connect its target to the commitment.

commitment
: A fresh random curve the signer publishes first, so that the challenge cannot be chosen after the response.

degree
: The size of an isogeny's kernel, the number of points that map to zero. An isogeny of degree 2^e is a path of e steps of degree 2.

dimension (of the embedding)
: The number of elliptic curves whose product carries the verifier's chain: two in SQIsign2D-West and the round-3 submission, four in SQIsignHD and here. Each step in dimension g acts on 2^g coordinates.

endomorphism ring
: All isogenies from a curve to itself. It is the secret: the security of every scheme here rests on its being hard to compute for a random supersingular curve.

isogeny
: A map between elliptic curves that respects their group law. It is determined by its kernel; "walking" from one curve to another means composing isogenies of small degree.

Kani embedding
: Kani's lemma: an isogeny of degree q between elliptic curves, together with one of degree $N - q$, sits inside an isogeny of degree N between products of curves. In dimension 4 the second isogeny is replaced by an endomorphism of the same curves built from a sum of two squares, $N - q = a_1^2 + a_2^2$, so nothing extra is computed or sent.

KAT
: Known-answer test: the reference's published keys and signatures for fixed seeds, which an implementation must reproduce byte for byte.

megacycle
: One million processor cycles, read with `rdtsc`; the unit of every cost in this paper, as in the reference's own benchmark.

response
: The isogeny from the commitment curve to the challenge curve that the signer, and only the signer, can compute. In dimension 4 it is described by its degree q and its action on a few points.

supersingular curve
: An elliptic curve over $\mathbb{F}_{p^2}$ whose endomorphism ring is a maximal order in a quaternion algebra. All curves in this paper are supersingular.

theta model
: The coordinate system the verifier's chain runs in. A point of a dimension-g variety has 2^g theta coordinates: four in dimension 2, sixteen in dimension 4.

torsion
: The points of a curve whose order divides a given number. The 2^r-torsion is where the response's action is written down; 2^r divides $p+1$ on these primes, so those points are defined over $\mathbb{F}_{p^2}$.

3 Related work

SQIsignHD The construction is due to Dartois, Leroux, Robert and Wesolowski [**?**]. The response has degree $q < 2^e$ with $2^e - q$ a prime congruent to 1 modulo 4, and the verifier rebuilds it as a $(2^e, 2^e, 2^e, 2^e)$-isogeny on $E_{\mathrm{com}}^2 \times E_{\mathrm{chl}}^2$, computed as two half-chains that meet in the middle (their Section 4). Their library, version 2.0, consists of a C signer, described in its README as a light modification of the SQIsign2D-West implementation, and a SageMath verifier built on the Theta dim4 package [**?**, **?**]. Its parameter script [**?**] sets $e = \lambda + \lceil \log_2 2\lambda \rceil$ and $r = \lceil e/2 \rceil + 2$ for primes of about 2λ bits, the size of the SQIsign primes before the attack of [**?**]; Section 4 explains why the rule for e does not carry over to the round-3 primes. The dimension-4 arithmetic, chains of 2-isogenies in the level-2 theta model, is Dartois's [**?**].

The submission The round-3 submission is a dimension-2 scheme in the SQIsign2D-West design [**?**, **?**]. Its reference code, the-sqisign [**?**], has a portable and an assembly-backed field arithmetic, and Chapter 5 of its specification fixes the parameters of our dimension-2 baseline.

sqisign-rs Our crate implements round-3 SQIsign in Rust with three formats: the specification's, a compressed format of its own that drops one matrix entry and recovers it from two Weil pairings, and, at level I, the dimension-4 compact format of this paper [**?**].

MIKE MIKE [**?**], a non-interactive key exchange built on Robert's module action [**?**], is the other recent system whose main cost is a chain of dimension-4 2-isogenies in the theta model. Its fourfolds carry an automorphism $[i]$ of order 4 that the chain respects, which it uses to compute each step from two points above the kernel instead of four [**?**, Appendix E.2]; and because its domain is a Weil restriction, the chain after the gluing is defined over $\mathbb{F}_p$ rather than $\mathbb{F}_{p^2}$. At level I it computes a chain of length 228, including the kernel computation and the gluing, in 16.0 megacycles in constant-time C with assembly field arithmetic [**?**, Table 1]. MIKE also argues that the round-3 primes are conservative, since the attack of [**?**] finds each candidate as a root of a large polynomial, a cost the specification's model does not count, and proposes $\log_2 p = 317$, 484 and 651 against 325, 504 and 668 [**?**, Section 4.5]. We keep the specification's primes, so that our figures compare directly with the round-3 submission.

Module representations Page, Robert and Soumier extend the Deuring correspondence to higher dimension [**?**]: maximal supersingular principally polarized abelian varieties correspond to unimodular Hermitian modules over a maximal quaternion order, and polarized isogenies to similitudes of such modules. On this basis they solve the principal ideal problem in matrix rings over quaternion orders, give a higher-dimensional Clapotis algorithm that makes the module representation explicit, and improve the bound of the KLPT^2 algorithm. We use their dictionary in Appendix A to phrase the descent of an automorphism along the verifier's chain.

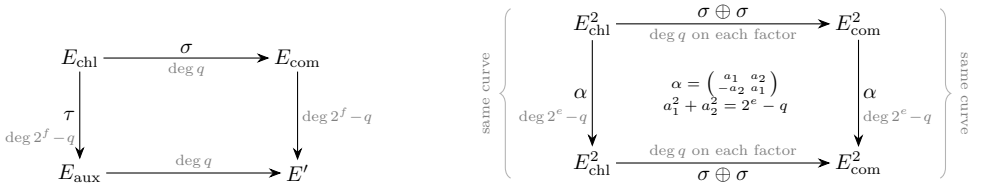
What we did not find To our knowledge the code measured here is the only implementation of a dimension-4 SQIsign signer and verifier in one codebase; the SQIsignHD authors' release is a C signer and a SageMath verifier for their primes. A GitHub search on 2026-09-26 for "SQIsignHD", "SQISignHD" and "Theta dim4" returned their two repositories and nothing else. We did not search further; an implementation we have missed would change this paragraph and nothing else in the paper.

4 Dimension 4

The cost The compact format's verifier runs a chain of dimension-4 isogeny steps, each on sixteen theta coordinates instead of the four of a dimension-2 step, 87 steps from each end at level I. It costs 6.4 times the dimension-2 verifier of the same crate at the same prime (85.4 against 13.3 megacycles), or 4.6 times with the two-point verifier of Section 7 (60.6 megacycles; Section 5).

What it buys In dimension 2 the verifier cannot run the response σ of degree q alone: Kani's embedding needs a second isogeny of degree $2^f - q$ from the challenge curve, the auxiliary isogeny τ, which the signer computes and whose target E_{aux} it sends (Figure 1, left). In dimension 4 the second isogeny is replaced by two integers with $a_1^2 + a_2^2 = 2^e - q$, which exist when $2^e - q$ is a prime congruent to 1 modulo 4 and which the verifier finds from q alone by Cornacchia's algorithm. The chain runs on $E_{\text{com}}^2 \times E_{\text{chl}}^2$ and involves no other curve (Figure 1, right). Dimension 4 removes the auxiliary curve, not the auxiliary freedom: another pair (a_1', a_2') with the same sum of squares, or another isogeny of degree q from the same curve, still yields a valid signature.

The signature carries the commitment curve, the degree q, and the action of σ on a basis of the 2^r-torsion: a 2×2 matrix modulo 2^r, of which three entries are sent and



(a) dimension 2: τ is an isogeny to a new curve; the signer computes it and E_{aux} is sent

(b) dimension 4: the auxiliary α is a pair of integers computed by the verifier from q; there is no auxiliary curve, so none is sent

Figure 1: The two ways a verifier rebuilds the response; both are Kani squares. (a) In dimension 2 the response σ of degree q is completed by an auxiliary isogeny τ of degree $2^f - q$ to a new curve; the signer computes τ and the signature carries E_{aux}. (b) In dimension 4 the same square sits on the doubled curves and the complement is the endomorphism $\alpha = \left(\begin{smallmatrix} a_1 & a_2 \\ -a_2 & a_1 \end{smallmatrix}\right)$ with $a_1^2 + a_2^2 = 2^e - q$, which the verifier computes from q; it acts on E^2_{chl} and on E^2_{com} alike, so each column of the square is one curve and there is no auxiliary curve to send. The signature carries E_{com}, q and the action of σ on the 2^r-torsion.

the fourth follows from the Weil pairing, $ad - bc \equiv kq \pmod{2^r}$ with k fixed by the two bases [**?**, Section 6.1]. At level I that is $2 \cdot 41 = 82$ bytes of curve, 22 bytes of q, three 12-byte scalars and two one-byte basis hints: 142 bytes, against 200 for the specification's format, which carries the auxiliary curve and a matrix modulo 2^{198}. The challenge is not sent: as in the round-3 specification, the verifier recomputes it by hashing the public key, the commitment curve and the message.

Which entry is dropped Write the transmitted action as $M = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, so that $ad - bc \equiv kq \pmod{2^r}$. Both bases pair to roots of unity of exact order 2^r, so k is odd; q is odd too, so $\det M$ is odd. Recovering the dropped entry means inverting another entry modulo 2^r, which must therefore be odd, and no single entry is odd on every signature. The rule: if a is odd, the signer drops d and the verifier computes $d \equiv (kq + bc)\, a^{-1} \pmod{2^r}$; if a is even, then $bc \equiv -kq$ is odd, so b and c are both odd, and the signer drops c, which the verifier computes as $c \equiv (ad - kq)\, b^{-1} \pmod{2^r}$. The verifier reads the case from the parity of a, so the rule costs no bit. The relation fixes r bits of the matrix and the format spends them on one entry; sending all four entries and only the top $e - r$ bits of q, whose low r bits the relation would then determine, would cost the same, so the pairings have nothing more to give.

Packing the signature The layout above rounds every field up to whole bytes: at level I each scalar takes 12 bytes for 89 bits, and q takes 22 bytes for 174 bits, two of which are known in advance since $q \equiv 3 \pmod 4$. Writing the commitment curve's Montgomery coefficient as two $\lceil \log_2 p \rceil$-bit integers, $(q-3)/4$ in $e-2$ bits, the three scalars in r bits each and the two hints in 8 bits each, padded once at the end, takes $2\lceil \log_2 p \rceil + (e-2) + 3r + 16$ bits: 1107, 1690 and 2223, that is 139, 212 and 278 bytes at levels I, III and V, against 142, 214 and 280 (Table 1). If each hint is a deterministic function of its curve, as in the round-3 specification, the verifier, which builds both curves, can recompute the hints instead of reading them, for 137, 210 and 276 bytes, at the cost of searching for the bases instead of decoding them. The signatures measured in Section 5 use the byte-aligned layout; the bit-packed one changes only the encoding.

Parameters for the round-3 primes SQIsignHD's exponent e bounds the response degree, $q < 2^e$. Every ideal class contains an ideal of norm at most about $0.90\sqrt{p}$ [**?**, Lemma 12],

so 2^e must exceed $\sqrt{p}$, and the paper takes "a few bits over $\sqrt{p}$" as sufficient on heuristic grounds (their Section 4.2). The library's rule $e = \lambda + \lceil \log_2 2\lambda \rceil$ is phrased in terms of the security parameter because it was set for primes with $p \approx 2^{2\lambda}$, where it amounts to $\log_2 \sqrt{p}$ plus 10 to 13 bits. The round-3 level-I prime has 70 bits more than $2^{2\lambda}$, and there the same formula gives $2^e = 2^{136}$, below $2^{162.6}$, the bound under which Lemma 12 guarantees an ideal in every class: no response would exist. What transfers is the content of the rule, $\sqrt{p}$ plus about ten bits.

We take $e = \lceil (\log_2 p)/2 \rceil + 11 = 174/264/346$ and, by the library's rule for the torsion the two half-chains need, $r = \lceil e/2 \rceil + 2 = 89/134/175$ (Table 1). The margin of 11 bits lies within the 10 to 13 bits that the library's rule gives on the primes it was written for; SQIsignHD's own prime uses 15.3. Two checks bound it from below. First, the number of ideal vectors of norm below 2^e grows as $2^{2 \cdot \text{margin}}$: about 2^{22} candidates, against one good norm in a few hundred, so a good q exists with overwhelming probability. Second, Nakagawa and Onuki's conservative countermeasure for PRISM-id, whose response reveals an isogeny of large prime degree, is a degree of at least $4\lambda/3$ bits [**?**, Section 5.3]; transferred to e, that is 170.7/256.0/341.3 against 174/264/346, cleared at every level, whereas a margin of 7 bits would fail it at level I.

The degree The signer samples vectors of the response ideal below 2^e and keeps the first whose norm q is odd, congruent to 3 modulo 4, and has $2^e - q$ a probable prime (40 Miller–Rabin rounds). The verifier repeats the test on q and decomposes $2^e - q$ by Cornacchia's algorithm. The sampled norms are odd, so one candidate in two passes the congruence; if $2^e - q$ behaves as a random e-bit integer, the density of good q is then $1/(e \ln 2)$: one candidate in 121 at level I, of which 60 reach a primality test. Section 5 measures the loop.

What the verifier checks The verifier decodes the key and the signature; recomputes the challenge; completes the dropped scalar by the rule above; tests q for goodness and writes $2^e - q = a_1^2 + a_2^2$ by Cornacchia's algorithm; and builds the kernel of the dimension-4 isogeny on $E_{\text{com}}^2 \times E_{\text{chl}}^2$ from a_1, a_2 and the transmitted action [**?**, Lemma 4 and Section 4.4]. It then computes the two half-chains of $\lceil e/2 \rceil$ steps from both ends, and accepts when the two middle codomains agree and the image of a point through the chain is what the transmitted action says, the two conditions of SQIsignHD's Section 4.5. A signature fails on any decoding error, on a degree that is not good, on a chain that lands on a product of curves where it should not, or on either condition.

Whose choices SQIsignHD's: the response as an ideal of good norm and the goodness test; the half-chains of $\lceil e/2 \rceil$ steps from both ends, with $r = \lceil e/2 \rceil + 2$; the action modulo 2^r with one scalar recovered from the pairing; the two acceptance conditions; and the library's basis-from-hint convention, kept so that its verifier checks our signatures unmodified. Ours: the move to the round-3 primes, e and its margin; the round-3 commitment and 2λ-bit challenge, recomputed rather than sent; the wire format, with two hint bytes in place of the library's packed bits, and its bit-packed variant; and the signer's sampler, which draws lattice vectors of the response ideal and rejects on goodness.

5 Benchmarks

Method All measurements come from one session on one machine, an AMD EPYC 7B13 guest (16 vCPUs, 63 GiB; the cycle counter at 3049.994 MHz), on 2026-11-02. One binary measures the Rust code, compiled by `rustc` 1.99.1 in release mode with fat link-time optimization and no target-CPU flag; the code chooses its assembly field kernels at run time from CPUID. Each operation is timed with `rdtsc` and the wall clock around the

Table 1: Dimension-4 parameters at the round-3 primes. Level I is built and measured; levels III and V are derived only.

	NIST I	NIST III	NIST V
p[a]	$3 \cdot 2^{324} - 1$	$27 \cdot 2^{500} - 1$	$17 \cdot 2^{664} - 1$
$\log_2 p$[a]	325.58	504.75	668.09
λ, challenge bits[a]	128	192	256
e, response bound $q < 2^e$[b]	174	264	346
$r = \lceil e/2 \rceil + 2$, transmitted torsion bits[c]	89	134	175
half-chain steps $\lceil e/2 \rceil$[c]	87	132	173
public key, bytes, predicted[d]	84	130	170
signature, bytes, predicted[d]	142	214	280
signature, bytes, bit-packed[e]	139	212	278
public key, bytes, measured	84	not built	not built
signature, bytes, measured	142	not built	not built
round-3 standard signature, bytes[a]	200	306	406

[a] The SQIsign round-3 specification [?].
[b] Ours: $e = \lceil (\log_2 p)/2 \rceil + 11$ (Section 4).
[c] SQIsignHD's rule for the torsion the half-chains need [?, Section 4.3].
[d] Field-by-field sums of the byte-aligned wire format (Section 4).
[e] $2\lceil \log_2 p \rceil + (e-2) + 3r + 16$ bits, padded once (Section 4); 137 / 210 / 276 without the hints.

Table 2: Public key + signature = total, in bytes, at the round-3 primes. Compact formats are level I only.

format	NIST I	NIST III	NIST V
standard (specification)	$83 + 200 = 283$	$129 + 306 = 435$	$169 + 406 = 575$
compressed (this crate)[a]	$83 + 174 = 257$	$129 + 267 = 396$	$169 + 352 = 521$
compact, dim. 4 (this crate)	$84 + 142 = 226$	not built	not built
compact, dim. 4, bit-packed[b]	$84 + 139 = 223$	$130 + 212 = 342$	$170 + 278 = 448$

[a] The compressed format's current layout. The session of Section 5 measured the earlier layout of 176 / 269 / 353 bytes.
[b] Predicted from the layout of Section 4, not observed in the session; levels III and V use the predicted public keys of Table 1.

call, on one pinned core, in four rounds that visit every block in turn; the tables report medians over 52 runs for key generation and signing in dimension 2, 200 for verification, and, for the compact format, 52 for key generation, 100 for signing and 200 for each verifier. Verification is measured from bytes: decode the key, decode the signature, verify.

Sizes Table 2 lists the sizes on the wire. At level I the compact key and signature take 226 bytes and 223 bit-packed, against 283 for the specification's format: a saving of 29 % on the signature, or 30.5 % bit-packed. The compressed dimension-2 format, with a 174-byte signature, sits between them.

Cycles Table 3 gives the session in megacycles. At level I the dimension-4 verifier costs 6.4 times the dimension-2 verifier of the same crate with four kernel points and 4.6 times with two, and dimension-4 signing costs 2.9 times dimension-2 signing, because the good degree is found by rejection (Table 5). Key generation is the same in both formats up to the basis convention (40.8 against 38.5 megacycles); the compressed format's recovery adds 16 % to a verification.

Milliseconds Table 4 gives wall-clock times at level I on two machines. Neither is what a deployment runs; the ratio between the rows is the number to carry.

Table 3: Megacycles at the round-3 primes, medians of the session. The last column is the dimension-4 verifier over the dimension-2 verifier of the same code.

implementation	level	keygen	sign	verify	× dim. 2
Rust, dimension 2	I	38.5	112.5	13.3	
Rust, dimension 2	III	142.2	392.9	33.7	
Rust, dimension 2	V	257.0	787.5	80.7	
Rust, dimension 2, compressed verify	I			15.4	
Rust, dimension 4 (compact)	I	40.8	329.0	85.4	6.4
Rust, dim. 4, two points, checked	I			60.6	4.6
Rust, dim. 4, two points, unchecked	I			59.6	4.5

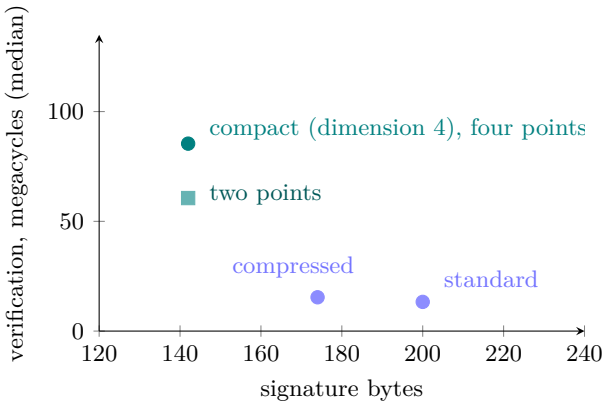


Figure 2: The trade at level I on the round-3 prime: signature bytes against verification cost for the three formats of one crate, the compact one with four and with two kernel points, from the session of Table 3. The compressed format is placed at its current size of 174 bytes; its verification cost was measured with the earlier 176-byte layout.

The rejection loop Table 5 compares the compact signer's search for a good degree over the session's signatures with the expectation of Section 4. A candidate costs one norm computation; one in two reaches a Miller–Rabin test on an e-bit number, and composites fail at the first round. This loop accounts for the spread of signing times: the median signature needs 79.5 samples, the mean 111.6, and the range is 2 to 523.

Memory At its peak, one compact verification at level I allocates 1674 KiB (1.64 MiB) of heap, measured with a counting allocator around the call in a session of 2026-09-26; the dimension-2 verifier allocates nothing.

6 Strong unforgeability and ECLIPSE

A signature scheme is strongly unforgeable when no one can produce a second valid signature on a message already signed. The SQIsign family is not: in the dimension-2 format the auxiliary isogeny can be replaced by another of the same degree, giving a second signature valid under the same key, as the round-3 specification states in its Section 8.4.2 [**?**]; the mechanism is written out in [**?**]. Dimension 4 removes the auxiliary curve from the wire but not the property: another isogeny of the same degree from the same commitment curve, with its own action on the torsion, is still a valid second signature. This is a property of the scheme, not a finding of this paper.

Table 4: Milliseconds at level I on the round-3 prime, medians, on the EPYC 7B13 guest of Section 5 (megacycles over its counter rate of 3049.994 MHz) and on an M4 Pro. The M4 Pro column is pending a run on that machine.

format	operation	EPYC (ms)	M4 Pro (ms)
dimension 2 (standard)	key generation	12.6	pending
	signing	36.9	pending
	verification	4.4	pending
	verification, compressed	5.0	pending
dimension 4 (compact)	key generation	13.4	pending
	signing	107.9	pending
	verification	28.0	pending

Table 5: The compact signer's good-degree loop at level I: measured over the session's signatures against the density $1/(e \ln 2)$ for odd q (and the figure for a candidate of random parity, which the sampler does not produce).

	measured	expected
signatures measured	100	
response samples per signature, mean	111.6	121
the same if q had random parity	???	241
response samples per signature, median (min, max)	79.5 (2, 523)	
primality tests per signature, mean	54.8	60
signing, Mcycles, median (min, max)	329.0 (48.8, 1900.9)	
signing, dimension 2, Mcycles, median	112.5	

A construction that is strongly unforgeable by design exists. Next to their salted scheme, the PRISM authors describe a variant whose signature is a single isogeny of prime degree $2^{2e-1} - s^2$ from the public-key curve, and state that the EUF-CMA proof of the salted scheme applies to a salted version of it [**?**, Section 5 and Remark 11]; we call that variant strong-PRISM. Our instantiation of it on the round-3 primes, ECLIPSE[1], is the subject of a forthcoming paper.

Dimension 4 buys strong-PRISM nothing. A PRISM signature carries no auxiliary curve in any dimension, so there is nothing to remove; a dimension-4 instantiation would have the same layout within a few bytes, and its hash target, a prime q with $2^a - q$ prime and congruent to 1 modulo 4, needs two primality conditions at once, a far sparser target than the single prime of the dimension-2 variant. A feasibility study for prism-rs [**?**] measured both and declined the dimension-4 route.

7 Conclusion

SQIsignHD's dimension-4 response carries over to the round-3 SQIsign primes once its response bound is restated in terms of $\sqrt{p}$ rather than the security parameter, and it keeps its defining advantage there: with no auxiliary curve to send, a level-I signature takes 142 bytes as built and 139 bit-packed, against 200 for the specification's format, and key and signature together take 226 bytes, or 223, the smallest level-I combination we are aware of at parameters that stand after the 2026 attack.

The verifier pays for it. In the same code, a dimension-4 verification costs 6.4 times a dimension-2 one at level I (85.4 against 13.3 megacycles), 4.6 times with the two-point

[1] ECLIPSE stands for Encoding-Canonical Large-prime Isogeny Signature Extension. The underlying construction, which we call strong-PRISM, is Section 5 of [**?**] with the salt of Section 6 as suggested in Remark 11; the authors did not name it.

verifier below, and needs about 1.6 MiB of heap where the dimension-2 verifier needs none; signing costs 2.9 times as much, because of the search for a good degree. The compact format therefore suits settings where bandwidth or storage dominates and a verification of a few tens of milliseconds is acceptable; the specification's dimension-2 format remains the better general-purpose choice. Neither format makes SQIsign strongly unforgeable; that requires a different construction, such as the PRISM variant of Section 6.

Two-point verification The verifier's chain respects an automorphism of order 4, as MIKE's does. On E^2 the automorphism $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ satisfies $J^2 = -1$ and $\alpha = a_1 + a_2 J$, so both α and $\sigma \oplus \sigma$ commute with J acting on E^2_{chl} and E^2_{com} alike (Appendix A). In every test we ran, the resulting automorphism of the fourfold acts on the theta coordinates our verifier uses as one fixed permutation of the sixteen coordinates, at every step of both half-chains, and maps two of the four kernel generators onto the other two. We prove that both facts hold at every step from the gluing on, for every input, honest or not, for the algorithm the verifier implements (Theorem 2); that our implementation computes that algorithm's theta structures is checked only in part (Remark 3). A verifier can therefore double and push only two kernel points along each half-chain and obtain the other two by permuting coordinates. We have implemented this variant as an option of our Rust port; it checks the condition at the gluing on each signature and falls back to four points otherwise. In the session of Section 5, two-point verification with the check costs 60.6 megacycles at level I against 85.4 for the four-point verifier, 0.71 of it, and 59.6 without the check, 0.70. In the port's tests, the two-point and the four-point verifier return the same verdict and the same rejection reason on the five level-I vectors and on 70 distinct tampered inputs, and the same codomain null point at each of the 844 plain steps of both half-chains over the five vectors, in both two-point modes.

Limitations There is only one implementation, and no independent round-3 dimension-4 signer to compare against. The SQIsignHD authors' verifier, re-parameterized to our prime and otherwise unmodified, accepts our signatures and rejects tampered ones [**?**], which shows that it agrees with our verifier on our signer's output and nothing more. Levels III and V are derived in Table 1 but not built, and the bit-packed layout of Section 4 is specified and sized but not implemented. Constant-time execution was not attempted for the dimension-4 signer or verifier: the verifier's integer arithmetic on the degree is variable-time on public data, and SQIsign signing is variable-time by design in the reference as well. The security argument is SQIsignHD's with our parameters substituted and has not been re-examined; the existence of a good degree in every class is a heuristic in their paper and remains one here. The two-point verifier is tested at level I only, on five vectors and their tampered variants, and is not the default; its correctness for every input rests on the conformance of the implementation to the algorithm of Theorem 2, which is checked only in part (Remark 3). The format is experimental and not for production.

Outlook Each limitation above is also a next step. Beyond them, faster dimension-4 arithmetic would narrow the gap in verification cost, which decides whether the 58 bytes saved at level I are worth having, and there is reason to expect room. MIKE's chain costs about 0.07 megacycles per step at level I, against about 0.49 for our four-point verifier and 0.35 for the two-point one (85.4 and 60.6 megacycles over $2 \cdot 87$ steps). The comparison is loose, since MIKE works over $\mathbb{F}_p$ on a different processor and our verifier also evaluates a point for its image check, but it suggests that much of our 6.4-fold penalty lies in the implementation rather than in the dimension (measured below: 1.18 of the counted floor at our own operation count). MIKE's computation of each codomain from two points and its faster doubling on the quotient by the automorphism [**?**, Appendix E.2] would save more than our two-point verifier, and its observation that additions, not multiplications,

are a significant share of the cost of dimension-4 theta arithmetic [**?**, Section 6] points toward field representations and vectorization tuned to the Hadamard transform.

In our two-point verifier the plain dimension-4 steps are 67.1 percent of the cost (point pushes 44.9 percent, codomain computations 22.2 percent), the setup before the chain 19.6 percent, the dimension-2 gluing chain 0.9 percent, the image check 6.5 percent and the gluing 6.0 percent, measured in the session of Section 5 (basis generation 7.48 megacycles, Cornacchia and kernel setup 4.23, dimension-2 steps 0.52, gluing 3.60, point pushes 26.88, codomain computations 13.28, image check 3.87; the whole call 60.37); the codomain computations, 33.1 percent of the plain steps, are the part a codomain-from-two-points step would address.

The two-point verifier's operation count, taken with counters in the field arithmetic at the session's commit, is 132 921 multiplications, 91 426 squarings, 435 363 additions, 54 inversions and 11 square roots in $\mathbb{F}_{p^2}$ per verification, 51.0 megacycles at the session's per-operation costs; at the reference count of our floor table, which takes a hand count of the SQIsignHD library's code where that count is below ours and our own count elsewhere, the same costs give 50.8. Measured in the same session, the verifier runs at 1.18 of the former floor and 1.19 of the latter, 9.6 megacycles above the latter, 5.8 of them in the two plain-step phases (1.15 for the point pushes, 1.21 for the codomains; 1.12 for the basis, 1.16 for the setup, 1.35 for the gluing, 1.16 for the image check). What remains is the sixteen-coordinate arithmetic of the plain steps on general points: their operation count is the library's, and the gap sits inside the Hadamard transform and the squaring kernels.

Three years separate the break of SIDH from a two-dimensional SQIsign in a NIST round, and one more separates that from primes moved by a new attack on the isogeny-path problem. Every construction in this paper is young, ours included, and the numbers describe code as it stood in November 2026, not a settled design; that is the reason for caution in using any of it.

A The order-4 symmetry of the verifier's chain

This appendix proves that the verifier's chain respects an automorphism of order 4 and that, for the algorithm the verifier implements, the automorphism acts on the theta coordinates of every fourfold from the gluing on as one fixed permutation (Theorem 2). This is what lets the two-point verifier of Section 7 apply MIKE's saving of two kernel points out of four [**?**, Appendix E.2] to the verifier of Section 4. Section A.7 separates what is established about our implementation from what remains unproven.

Notation Throughout, E denotes a supersingular elliptic curve over $\mathbb{F}_{p^2}$ and $E^2 = E \times E$. Homomorphisms between squares of curves are written as 2×2 matrices whose entries are homomorphisms between the curves, acting on column vectors:

$$\begin{pmatrix} u & v \\ w & z \end{pmatrix} \begin{pmatrix} P \\ Q \end{pmatrix} = \begin{pmatrix} u(P) + v(Q) \\ w(P) + z(Q) \end{pmatrix}.$$

An integer n stands for the multiplication map $[n]$. For a homomorphism ψ we write $\widehat{\psi}$ for its dual. With respect to the product polarizations, the dual of a matrix is the transpose of the matrix of duals; in particular the dual of an integer matrix is its transpose. We write E_{chl} for the challenge curve, E_{com} for the commitment curve, $\sigma\colon E_{\mathrm{chl}} \to E_{\mathrm{com}}$ for the response, of odd degree q, and a_1, a_2 for integers with $a_1^2 + a_2^2 = 2^e - q$, as in Section 4. Finally

$$A_1 = E_{\mathrm{chl}}^2, \qquad A_2 = E_{\mathrm{com}}^2, \qquad A = A_1 \times A_2,$$

so that A is the abelian fourfold on which the verifier's chain starts.

A.1 The automorphism J of E^2

Definition 1. On any square E^2, let

$$J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \text{that is,} \qquad J(P,Q) = (Q,-P),$$

and on $A = A_1 \times A_2$ let $D = \mathrm{diag}(J,J)$ be J acting on each square separately.

Lemma 1.

(i) $J^2 = -1$, *so* J *has order 4.*

(ii) The endomorphism $\alpha = \left(\begin{smallmatrix} a_1 & a_2 \\ -a_2 & a_1 \end{smallmatrix}\right)$ *of Figure 1 equals* $a_1 + a_2 J$, *and its dual is* $\widehat{\alpha} = \alpha^{\mathsf{T}} = a_1 - a_2 J$.

(iii) $\widehat{\alpha}\,\alpha = \alpha\,\widehat{\alpha} = a_1^2 + a_2^2 = 2^e - q$.

(iv) $\widehat{J} = J^{\mathsf{T}} = -J = J^{-1}$, *so* J *preserves the product polarization of* E^2*: it is an automorphism of* E^2 *as a principally polarized abelian surface.*

Proof. (i) $J^2(P,Q) = J(Q,-P) = (-P,-Q)$. (ii) We have

$$(a_1 + a_2 J)(P,Q) = \big([a_1]P + [a_2]Q,\ -[a_2]P + [a_1]Q\big),$$

which is $\alpha(P,Q)$; the dual of an integer matrix is its transpose, and $J^{\mathsf{T}} = -J$. (iii) $(a_1 + a_2 J)(a_1 - a_2 J) = a_1^2 - a_2^2 J^2 = a_1^2 + a_2^2$, since J commutes with integers. (iv) $J^{\mathsf{T}} J = 1$ by direct computation, and an automorphism γ preserves the polarization exactly when $\widehat{\gamma}\gamma = 1$. □

A.2 The verifier's isogeny commutes with D

Write $\Sigma = \mathrm{diag}(\sigma,\sigma)\colon A_1 \to A_2$, the map $\sigma \oplus \sigma$ of Figure 1, so that $\widehat{\Sigma} = \mathrm{diag}(\widehat{\sigma},\widehat{\sigma})$. Consider the Kani endomorphism [**?**, Section 4]

$$F = \begin{pmatrix} \alpha & \widehat{\Sigma} \\ -\Sigma & \widehat{\alpha} \end{pmatrix} : A_1 \times A_2 \longrightarrow A_1 \times A_2, \qquad F(x_1,x_2) = \big(\alpha(x_1) + \widehat{\Sigma}(x_2),\ -\Sigma(x_1) + \widehat{\alpha}(x_2)\big),$$

where the α in the top-left corner acts on A_1 and the $\widehat{\alpha}$ in the bottom-right corner on A_2. Its dual is

$$\widehat{F} = \begin{pmatrix} \widehat{\alpha} & -\widehat{\Sigma} \\ \Sigma & \alpha \end{pmatrix}.$$

The verifier's dimension-4 isogeny is this endomorphism with the roles of the two curves exchanged: $(E_{\mathrm{chl}}, E_{\mathrm{com}}, \sigma)$ replaced by $(E_{\mathrm{com}}, E_{\mathrm{chl}}, \widehat{\sigma})$, so that it acts on $E_{\mathrm{com}}^2 \times E_{\mathrm{chl}}^2$ (Lemma 11 gives it explicitly). Nothing below depends on which curve is which, so every statement about F and $\widehat{F}$ applies to it.

Lemma 2. $\widehat{F}F = F\widehat{F} = 2^e$. *Hence* F *is an isogeny of degree* 2^{4e}, *and* $\ker F = \widehat{F}\big(A[2^e]\big)$.

Proof. Every entry of F and $\widehat{F}$ is an integer matrix or a componentwise map $\mathrm{diag}(\psi,\psi)$, and these commute with each other: for an integer matrix M and any homomorphism ψ of curves, $\mathrm{diag}(\psi,\psi)M = M\,\mathrm{diag}(\psi,\psi)$, because ψ is a group homomorphism. The diagonal blocks of $\widehat{F}F$ are $\widehat{\alpha}\alpha + \widehat{\Sigma}\Sigma = (2^e - q) + q = 2^e$ and $\Sigma\widehat{\Sigma} + \alpha\widehat{\alpha} = 2^e$, by Lemma 1(iii) and $\widehat{\sigma}\sigma = \sigma\widehat{\sigma} = q$; the off-diagonal blocks are $\widehat{\alpha}\widehat{\Sigma} - \widehat{\Sigma}\widehat{\alpha} = 0$ and $\Sigma\alpha - \alpha\Sigma = 0$. The same holds for $F\widehat{F}$. Then $(\deg F)^2 = \deg[2^e] = 2^{8e}$ on a fourfold, so $\deg F = 2^{4e}$. Finally $\widehat{F}(A[2^e]) \subseteq \ker F$ because $F\widehat{F} = 2^e$, and both groups have 2^{4e} elements, since $\ker \widehat{F} \subseteq A[2^e]$ has 2^{4e} elements and $|A[2^e]| = 2^{8e}$. □

Proposition 1. *$F \circ D = D \circ F$. In particular* $\ker F$ *is stable under D.*

Proof. Since $D = \mathrm{diag}(J, J)$, the identity $FD = DF$ holds block by block exactly when each block of F commutes with J. The blocks α and $\widehat{\alpha}$ are polynomials in J (Lemma 1(ii)), and Σ, $\widehat{\Sigma}$ are componentwise maps, which commute with every integer matrix (proof of Lemma 2). Explicitly, $J\Sigma(P, Q) = J(\sigma P, \sigma Q) = (\sigma Q, -\sigma P) = \Sigma(Q, -P) = \Sigma J(P, Q)$. If $F(x) = 0$ then $F(Dx) = DF(x) = 0$. The blocks of $\widehat{F}$ are of the same kinds, so $\widehat{F}D = D\widehat{F}$ as well. □

A.3 Consequences along the chain

The verifier computes F as a chain of e isogenies of degree 2. Write $K = \ker F$ and, for $0 \le k \le e$, let $F_k \colon A \to A_k = A/[2^{e-k}]K$ be the partial chain after k steps, so that $F_e = F$ up to isomorphism. The two half-chains that meet in the middle are partial chains of this kind, one read from each end.

Proposition 2. *For every k, there is an automorphism D_k of A_k with $D_k \circ F_k = F_k \circ D$. It satisfies $D_k^2 = -1$ and $\widehat{D_k} = -D_k$, and it preserves the principal polarization of A_k.*

Proof. D is a group automorphism with $D(K) = K$ (Proposition 1), hence $D([2^{e-k}]K) = [2^{e-k}]K$, so $F_k \circ D$ vanishes on $\ker F_k$ and factors as $D_k \circ F_k$; D_k is an automorphism because D is. The identities $D_k^2 = -1$ and $\widehat{D_k} = -D_k$ follow from $D^2 = -1$ and $\widehat{D} = -D$ (Lemma 1) by uniqueness of the factorization. The principal polarization of A_k is the one induced from 2^k times that of A through the isotropic kernel $\ker F_k$; D preserves both, so D_k preserves the induced one. □

Remark 1 (The module viewpoint). The varieties here are maximal: each curve has $(p+1)^2$ points over $\mathbb{F}_{p^2}$. By the anti-equivalence $A \mapsto M_A = \mathrm{Hom}(A, E_0)$ of [**?**, Theorem 2.5], under which principal polarizations correspond to unimodular Hermitian forms H_A and isogenies to module monomorphisms [**?**, Table 1], the automorphism D corresponds to $f \mapsto f \circ D$ on M_A, which preserves H_A by Lemma 1(iv) and squares to -1. The partial chain F_k corresponds to the inclusion $\psi_k \colon M_{A_k} \hookrightarrow M_A$, $f \mapsto f \circ F_k$, whose image consists of the morphisms vanishing on $\ker F_k$. Since $D(\ker F_k) = \ker F_k$, that image is stable under $f \mapsto f \circ D$, and $f \circ D_k \circ F_k = f \circ F_k \circ D$ shows that D_k corresponds to the restriction of $f \mapsto f \circ D$ to it. This is Proposition 2 in module form: D descends along the chain exactly because each partial kernel, equivalently each image submodule, is D-stable.

Proposition 3. *Every A_k is of Weil type in the sense of [**?**, Definition 4.1]: D_k embeds $\mathbb{Z}[i]$ in* $\mathrm{End}(A_k)$*, with $i \mapsto D_k$, and acts on the tangent space at the origin with eigenvalues $i, i, -i, -i$.*

Proof. On the tangent space of E^2 the differential of J is the matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, whose eigenvalues are i and $-i$; these lie in $\mathbb{F}_{p^2}$ because $p \equiv 3 \pmod 4$. Hence $D = \mathrm{diag}(J, J)$ has eigenvalues $i, i, -i, -i$ on the tangent space of A. Isogenies of degree a power of 2 are separable in odd characteristic, so the differential of F_k is an isomorphism, and it intertwines D with D_k (Proposition 2); the eigenvalues are therefore the same. This is the argument of [**?**, proof of Proposition G.1]. □

Proposition 4. *For any matrix the verifier receives, honest or not, the kernel it builds is stable under D. Whenever that kernel defines a chain, that is, when it is maximal isotropic and the computation succeeds, the conclusions of Propositions 2 and 3 hold for that chain.*

Proof. The verifier builds its kernel from the transmitted action rather than from σ: the transmitted matrix defines a map σ' on the 2^r-torsion (the map τ of Section A.6, with the curves exchanged as in Section A.2), and the verifier uses $\Sigma' = \mathrm{diag}(\sigma', \sigma')$ in place of Σ.

Like Σ, the map Σ' acts componentwise, so it commutes with J, and the kernel generators $(\widehat{\alpha}(x), \Sigma'(x))$ satisfy $D(\widehat{\alpha}(x), \Sigma'(x)) = (\widehat{\alpha}(Jx), \Sigma'(Jx))$, which is again of that form. The proofs of Propositions 2 and 3 use only that the kernel is D-stable and that the chain has 2-power degree. □

A.4 The kernel as a module over $\mathbb{Z}[i]$

Let $R_e = (\mathbb{Z}/2^e)[i]$, acting on 2^e-torsion with i acting as J (or D).

Lemma 3. *$E^2[2^e]$ is a free R_e-module of rank 2, and so is* $\ker F$.

Proof. Let (S, T) be a basis of $E[2^e]$. Since $J(S, 0) = (0, -S)$ and $J(T, 0) = (0, -T)$, the four points $(S, 0), (T, 0), (0, S), (0, T)$, which form a $\mathbb{Z}/2^e$-basis of $E^2[2^e]$, are $(S, 0)$, $(T, 0)$, $-J(S, 0)$ and $-J(T, 0)$; so $(S, 0)$ and $(T, 0)$ form an R_e-basis. For the kernel, consider

$$\iota \colon A_1[2^e] \to A, \qquad \iota(x) = \widehat{F}(x, 0) = \big(\widehat{\alpha}(x), \Sigma(x)\big).$$

Its image lies in $\ker F$ by Lemma 2. It is injective, because $\widehat{\alpha}$ has odd degree $(2^e - q)^2$ and is therefore injective on 2-power torsion. Its domain has 2^{4e} elements, as does $\ker F$, so ι is a bijection onto $\ker F$. Finally ι commutes with J, so it is R_e-linear, and $\ker F \cong A_1[2^e] = E^2_{\mathrm{chl}}[2^e]$ is free of rank 2. □

MIKE uses the corresponding statement about its own kernel [**?**, proof of Corollary G.1] to show that the pairing e_{1+i} is alternate on its fourfolds, which in turn is what makes its adapted theta coordinates exist. On the starting fourfold A the analogous pairing is alternate.

Lemma 4. *The Weil–Cartier pairing e_{1+D} on $A[1 + D]$ is alternate.*

Proof. It suffices to treat one square E^2, since A is a product of two and the pairings multiply. If $(1 + J)(P, Q) = (P + Q, Q - P) = 0$, then $P = -Q$ and $2P = 0$, so $\ker(1 + J) = \{(u, u) : u \in E[2]\}$. With the formula of [**?**, proof of Corollary G.1], $e_{1+J}(x, x) = e_2(Jx_0, x_0)$ for any x_0 with $(1 + J)x_0 = x$. For $x = (u, u)$ take $x_0 = (0, u)$, since $(1 + J)(0, u) = (0, u) + (u, 0) = (u, u)$. Then $e_2(Jx_0, x_0) = e_2\big((u, 0), (0, u)\big) = e_2(u, 0)\, e_2(0, u) = 1$. □

A.5 The verifier's theta coordinates

This subsection proves how D acts in level-2 theta coordinates once it acts correctly on the basis inducing them; Section A.6 proves that it does, on the theta structures of the verifier's algorithm. Throughout, π is the permutation of $\{0, \dots, 15\}$ that exchanges bits 0 and 2 and bits 1 and 3 of the index $a = i_0 + 2i_1 + 4i_2 + 8i_3$; it is an involution. We identify a with its multi-index $\mathbf{i} = (i_0, i_1, i_2, i_3) \in \{0, 1\}^4$, write $\theta_{\mathbf{v}}$ for $\theta_{\mathbf{v} \bmod 2}$ when $\mathbf{v}$ is an integer vector, and $\langle \mathbf{i} \mid \mathbf{j} \rangle = \sum_l i_l j_l$.

Theta structures and bases Let A' be a principally polarized abelian fourfold over an algebraically closed field $\Bbbk$ of odd characteristic (the varieties of the chain are defined over $\mathbb{F}_{p^2}$, and we work over its algebraic closure, as does [**?**, Section 2]), let $\mathcal{L} = \mathcal{L}_0^2$ with $\mathcal{L}_0$ a symmetric line bundle representing the principal polarization, so that $\mathcal{L}$ is totally symmetric of type $(2, 2, 2, 2)$, and let ζ be a primitive fourth root of unity. By [**?**, Theorem 5], a symmetric level-2 theta structure for $\mathcal{L}$ is determined by a symplectic isomorphism from $(\mathbb{Z}/4)^4 \times (\mathbb{Z}/4)^4$ to $A'[4]$, that is, by the image $B = (s_0, \dots, s_3; t_0, \dots, t_3)$ of the ζ-canonical basis $(e_0, \dots, e_3; \chi_0, \dots, \chi_3)$, a ζ-symplectic basis of $A'[4]$. We call it the structure *induced by B* and write θ^B_a for its coordinates. Bit l of the index is the l-th coordinate of $(\mathbb{Z}/2)^4$ [**?**, Section 3.2], so it belongs to the pair (s_l, t_l). For a tuple $x = (x_0, \dots, x_3)$ of points and a

4×4 integer matrix U we write xU for the tuple $\left(\sum_l U_{lj} x_l\right)_j$, as for coefficient vectors, and for $M = \left(\begin{smallmatrix} U & V \\ W & U' \end{smallmatrix}\right)$ with 4×4 blocks, $BM = (sU + tW; sV + tU')$: the columns of M give the new basis in terms of the old.

Index convention In a chain the half t of the basis inducing the structure of each fourfold lies above the kernel of the next step [**?**, Section 4.1], and positions are kept from step to step: if the structure of A_k is induced by $(s;t)$ and the step f uses 8-torsion points T'' with $2T'' = t$, the structure of its codomain is induced by $(f(s); f(T''))$, position by position [**?**, Theorem 56], [**?**, Section 4.2]. On every fourfold of a half-chain, index bit l is therefore labeled by the l-th kernel generator g_l, which the verifier takes from column $4 + l$ of its starting matrix, in column order (Lemma 9). Remark 3 records that our crate follows this convention.

Lemma 5 (The product)**.** *Give both copies of a curve E the same level-2 theta structure, so that $\theta_{(i,j)}(P,Q) = \theta_i(P)\,\theta_j(Q)$ on E^2. Then $\theta_{(i,j)}(J(P,Q)) = \theta_{(j,i)}(P,Q)$: J exchanges the two index bits of the square, with no scalars.*

Proof. $\theta_{(i,j)}(Q,-P) = \theta_i(Q)\,\theta_j(-P) = \theta_j(P)\,\theta_i(Q)$, because level-2 theta functions are even. □

Lemma 6 (Change of structure)**.** *Let A', $\mathcal{L}$ and ζ be as above, B a ζ-symplectic basis of $A'[4]$ and $M = \left(\begin{smallmatrix} U & V \\ W & U' \end{smallmatrix}\right) \in \mathrm{Sp}_8(\mathbb{Z}/4)$. Then BM is a ζ-symplectic basis, and there is $\lambda \in \Bbbk^*$ such that, for all $\mathbf{i}$,*

$$\theta^{BM}_{\mathbf{i}} = \lambda \sum_{\mathbf{j} \in \{0,1\}^4} \zeta^{\langle \mathbf{i}|\mathbf{j}\rangle - \langle U\mathbf{i}+V\mathbf{j}+2\mathbf{i}_0 \,|\, W\mathbf{i}+U'\mathbf{j}\rangle}\, \theta^B_{U\mathbf{i}+V\mathbf{j}+\mathbf{i}_0}$$

as an identity of sections of $\mathcal{L}$, for every $\mathbf{i}_0$ such that the section $T_{\mathbf{i}_0} = \sum_{\mathbf{j}} \zeta^{-\langle V\mathbf{j}+2\mathbf{i}_0|U'\mathbf{j}\rangle}\theta^B_{\mathbf{i}_0+V\mathbf{j}}$ is nonzero; any two such $\mathbf{i}_0$ give the same coordinates up to a scalar. If $V = 0$, then:

(i) $U^{\mathsf{T}}U' \equiv 1 \pmod 4$, so $U' = U^{-\mathsf{T}}$ modulo 4;

(ii) $\mathbf{i}_0 = 0$ is admissible and is the only admissible index, whatever the theta null point;

(iii) $\theta^{BM}_{\mathbf{i}} = \lambda\,\zeta^{-\langle U\mathbf{i}|W\mathbf{i}\rangle}\,\theta^B_{U\mathbf{i}}$; in particular $\theta^{BM}_{\mathbf{i}} = \lambda\,\theta^B_{U\mathbf{i}}$ when also $W = 0$.

Proof. The hypotheses of [**?**, Theorem 12] are: $\mathcal{L}$ totally symmetric, here of type $\delta = (2,2,2,2)$; a symmetric pair of theta structures for $(\mathcal{L}, \mathcal{L}^2)$, which B provides by [**?**, Theorem 5]; ζ a primitive $2d_g$-th root of unity, here $d_g = 2$; and a symmetric automorphism ψ of the Heisenberg group whose matrix in the ζ-canonical basis is M, which exists because symmetric automorphisms surject onto $\mathrm{Sp}_8(\mathbb{Z}/4)$ [**?**, Section 2.11]. Composing with ψ maps the ζ-canonical basis to BM, so the new structure is the one induced by BM, and it depends on M only [**?**, Theorem 5(ii)]. With these identifications, and with Dartois's blocks $\left(\begin{smallmatrix} A & C \\ B & D \end{smallmatrix}\right)$ written $\left(\begin{smallmatrix} U & V \\ W & U' \end{smallmatrix}\right)$, Theorem 12 of [**?**] is the displayed formula, with $\langle\cdot|\cdot\rangle$ the standard product since all $d_l = 2$. Its condition on $\mathbf{i}_0$ is a condition on sections: the proof obtains $\theta'_0 = \lambda T_{\mathbf{i}_0}$ from [**?**, Proposition 8], for any $\mathbf{i}_0$ with $T_{\mathbf{i}_0} \neq 0$. Since θ' is determined by the new structure up to a scalar, the choice of admissible $\mathbf{i}_0$ does not matter.

(i) This is [**?**, Lemma 11] with $\Delta = 1$: $U^{\mathsf{T}}U' - W^{\mathsf{T}}V \equiv 1 \pmod 4$. (ii) With $V = 0$ we have $\zeta^{-2\langle \mathbf{i}_0|U'\mathbf{j}\rangle} = (-1)^{\langle U'^{\mathsf{T}}\mathbf{i}_0|\mathbf{j}\rangle}$, so $T_{\mathbf{i}_0} = \theta^B_{\mathbf{i}_0}\sum_{\mathbf{j}}(-1)^{\langle U'^{\mathsf{T}}\mathbf{i}_0|\mathbf{j}\rangle}$. By (i), U' is invertible modulo 2, so the sum is 16 for $\mathbf{i}_0 = 0$ and 0 otherwise. The sixteen θ^B_a form a basis of the sections of $\mathcal{L}$ [**?**, Section 2.4] and $16 \neq 0$ in odd characteristic, so $T_0 = 16\theta^B_0 \neq 0$. The condition concerns the section θ^B_0, not its value at the origin, so it holds even where the theta null point has $\theta^B_0(0) = 0$. (iii) With $V = 0$ and $\mathbf{i}_0 = 0$ the exponent is $\mathbf{i}^{\mathsf{T}}(1 - U^{\mathsf{T}}U')\mathbf{j} - \langle U\mathbf{i}|W\mathbf{i}\rangle$ for any integer lifts of $\mathbf{i}, \mathbf{j}$, and the first term vanishes modulo 4 by (i). The summand no longer depends on $\mathbf{j}$, and the factor 16 goes into λ. □

Lemma 7 (Transport)**.** *Let A', $\mathcal{L}$, ζ and B be as in Lemma 6, and let φ be an automorphism of A' preserving its principal polarization. Then $\varphi(B)$ is ζ-symplectic, and there is $c \in \Bbbk^*$ such that $\theta_a^{\varphi(B)}(\varphi(z)) = c\,\theta_a^B(z)$ for all $z \in A'$ and all a. Consequently, if $\varphi(B) = BM$ with $M = \mathrm{diag}(U, U')$, there is $\mu \in \Bbbk^*$ such that*

$$\theta_a^B(\varphi(z)) = \mu\,\theta_{U^{-1}a}^B(z) \qquad \text{for all } z \text{ and } a:$$

φ acts on the coordinates of the structure induced by B through the inverse of the matrix by which it acts on B.

Proof. Since φ preserves the polarization, it preserves the Weil pairing e_4, so $\varphi(B)$ is ζ-symplectic. Next, $\varphi^*\mathcal{L} \cong \mathcal{L}$. Indeed $\varphi^*\mathcal{L}_0$ is symmetric, because φ commutes with $[-1]$, and algebraically equivalent to $\mathcal{L}_0$, because φ preserves the polarization; so $\varphi^*\mathcal{L}_0 \cong t_x^*\mathcal{L}_0$ for some x with $2x = 0$, and then $\varphi^*\mathcal{L} \cong t_x^*(\mathcal{L}_0^2) \cong \mathcal{L}$, since $A'[2]$ is the kernel of the polarization of $\mathcal{L}_0^2$.

Fix an isomorphism $\beta\colon \varphi^*\mathcal{L} \to \mathcal{L}$. The pair (φ, β) is an instance of the situation $(*)$ of [**?**, p. 290], with $\pi = \varphi$ an isomorphism, so that its kernel and the corresponding level subgroup are trivial. By [**?**, Proposition 2, p. 291] it induces an isomorphism τ of the theta group of $\mathcal{L}$ lying over φ (Mumford's map (2) on p. 302, the map α_f of [**?**, proof of Theorem 56]), and $(\varphi, \beta^{\otimes 2})$ likewise induces τ_2 on the theta group of $\mathcal{L}^2$. Mumford states [**?**, p. 312, diagram (s), with $n = 2$], for every separable isogeny, that these maps commute with his maps ε_2 and η_2 [**?**, pp. 309–311]. They therefore commute with $\delta_2 = \eta_2 \circ \varepsilon_2$ [**?**, Proposition 5(iii), p. 311] on both theta groups, and hence with δ_{-1}, since $\delta_2(z) = z^3\delta_{-1}(z)$ [**?**, p. 312]. Write the compatible pair of symmetric theta structures induced by B as Mumford's isomorphisms j_1 and j_2 from the theta groups of $\mathcal{L}$ and $\mathcal{L}^2$ onto his standard groups of types $(2,2,2,2)$ and $(4,4,4,4)$ [**?**, p. 317]. By these commutations, $(j_1 \circ \tau^{-1}, j_2 \circ \tau_2^{-1})$ is again a compatible pair of symmetric theta structures in the sense of [**?**, p. 317]. Since τ_2 lies over φ, its symplectic isomorphism onto $A'[4]$ maps the ζ-canonical basis to $\varphi(B)$, so by [**?**, Theorem 5(ii)] $j_1 \circ \tau^{-1}$ is the structure induced by $\varphi(B)$. Finally, Mumford's isogeny theorem [**?**, Theorem 4, pp. 302–303], applied to (φ, β) with trivial kernel and with the structure $j_1 \circ \tau^{-1}$ on the target, so that the map σ there is the identity, shows that φ^* maps the section with coordinates f for the structure induced by $\varphi(B)$ to c times the section with coordinates f for the structure induced by B, for a constant $c \in \Bbbk^*$. In the form of [**?**, Theorem 3], this is $\theta_a^{\varphi(B)} \circ \varphi = c\,\theta_a^B$ for all a, that is, $\theta_a^{\varphi(B)}(\varphi(z)) = c\,\theta_a^B(z)$, which is the first assertion.

For the second, Lemma 6 with $V = W = 0$ gives $\theta_a^{BM} = \lambda\,\theta_{Ua}^B$. Hence $\theta_a^B(z) = c^{-1}\theta_a^{\varphi(B)}(\varphi(z)) = c^{-1}\lambda\,\theta_{Ua}^B(\varphi(z))$, and putting $b = Ua$, which runs over all indices because U is invertible modulo 2, $\theta_b^B(\varphi(z)) = c\lambda^{-1}\theta_{U^{-1}b}^B(z)$. □

As a check on the direction, consider E^2 with the product structure of Lemma 5, induced by $s = ((T_1, 0), (0, T_1))$ and $t = ((T_2, 0), (0, T_2))$ for a basis (T_1, T_2) of $E[4]$. Then $Js = sU$ and $Jt = tU$ with $U = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, and U^{-1} exchanges the two index bits, as Lemma 5 says. There U^{-1} and U induce the same permutation, as they will for the matrix U of Lemma 13 below, whose reduction modulo 2 is an involution; the inverse matters only for permutations of larger order.

Lemma 8 (One step)**.** *Let $f\colon A_k \to A_{k+1}$ be a step of the chain and U a signed permutation matrix, so that $U^{-\mathsf{T}} = U$. Suppose that D_k acts on the basis $(s; t)$ inducing the structure of A_k by $\mathrm{diag}(U, U)$ modulo 4, and that the 8-torsion points T'' with $2T'' = t$ used by the step satisfy $D_kT'' = T''U$ exactly, as points. Then D_{k+1} acts on the basis $(f(s); f(T''))$, which induces the structure of A_{k+1}, by the same block matrix modulo 4.*

Proof. By Proposition 2, $D_{k+1} \circ f = f \circ D_k$. The hypothesis on s is an identity between points of $A_k[4]$, so $D_{k+1}f(s) = f(D_ks) = f(s)\,U$; and $D_{k+1}f(T'') = f(D_kT'') = f(T'')\,U$ exactly. □

The 8-torsion points T'' of the hypothesis are the chain's kernel generators, doubled and pushed to step k. Since every step commutes with D, they inherit an exact pairing $Dg_0 = \pm g_2$, $Dg_1 = \pm g_3$ from the generators on A, if the generators have it.

Theorem 1 (Conditional on the base case)**.** *Let U be a signed permutation matrix whose reduction modulo 2 is the permutation matrix of $(0\ 2)(1\ 3)$. Suppose that, for one half-chain, the four kernel generators on A satisfy $Dg_0 = \pm g_2$ and $Dg_1 = \pm g_3$ as points, and that D acts on the basis inducing the theta structure of the gluing codomain by $\mathrm{diag}(U, U)$ modulo 4. Then on every fourfold of that half-chain from the gluing codomain on, D_k acts on the level-2 theta coordinates as π, that is, $\theta_a(D_k z) = \mu_k\, \theta_{\pi(a)}(z)$ for all z and a, and the two-point chain of Section 7 computes the same kernels, isogenies and codomains as the four-point chain.*

Proof. Lemma 8, with the pairing of the pushed generators noted above, carries the block action from the gluing codomain to every later fourfold of the half-chain. On each of them, Lemma 7 turns it into $\theta_a(D_k z) = \mu_k\, \theta_{U^{-1}a}(z)$. Modulo 2, $U^{-1} = U^{\mathsf{T}}$ is the permutation matrix of $(0\ 2)(1\ 3)$, so by the index convention $U^{-1}a = \pi(a)$. The step from A_k uses the 8-torsion points $T''_0, \ldots, T''_3$ above the kernel, and $T''_2 = \pm D_k T''_0$, $T''_3 = \pm D_k T''_1$. Since level-2 theta functions are even, $\theta(T''_2) = \theta(D_k T''_0) = \mu_k\, \pi(\theta(T''_0))$, and likewise for T''_3. The two-point chain uses $\pi(\theta(T''_0))$ and $\pi(\theta(T''_1))$ in place of the theta coordinates of T''_2 and T''_3: the same projective points, hence the same kernel, isogeny and codomain, and the doublings and images it computes for T''_0, T''_1 are those of the four-point chain. □

Lemma 9 (The structured kernel block)**.** *In the verifier's coefficient layout, where row r of an 8×8 matrix is the coefficient of the base point $\lfloor r/4 \rfloor$ on copy $r \bmod 4$, with copies $0, 1$ on E_{com} and $2, 3$ on E_{chl}, D acts on coefficient vectors by $(v_0, v_1, v_2, v_3) \mapsto (v_1, -v_0, v_3, -v_2)$ on each half. On the kernel block of the starting matrix (the columns $(a_1, a_2, 1, 0)$, $(-a_2, a_1, 0, 1)$ in the first half and their analogs with q in the second), this maps the first column to minus the third and the second to minus the fourth.*

Proof. $(a_1, a_2, 1, 0) \mapsto (a_2, -a_1, 0, -1) = -(-a_2, a_1, 0, 1)$, and likewise for the columns supported on the second half. □

The verifier's generators are not this block, however: they are columns of matrices obtained from it through generic symplectic completions, as the next subsection explains.

A.6 The base case

This subsection proves the hypotheses of Theorem 1 for the verifier's algorithm, on the theta structures that algorithm uses (Definition 2): for the verifier's own starting matrices, whatever symplectic completion it uses, and for every input for which its half-chains are defined. Whether our implementation computes those structures is a separate question, taken up in Section A.7.

Base points Let (P, Q) be the basis of $E_{\mathrm{chl}}[2^r]$ and (R, S) the basis of $E_{\mathrm{com}}[2^r]$ that the verifier uses, and k the integer with $e(R, S) = e(P, Q)^k$, where e is the Weil pairing on 2^r-torsion; k is odd because both pairings have exact order 2^r (Section 4). The verifier reads the transmitted matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$, completed by the rule of Section 4, as the homomorphism $\tau\colon E_{\mathrm{com}}[2^r] \to E_{\mathrm{chl}}[2^r]$ with $\tau(R) = aP + bQ$ and $\tau(S) = cP + dQ$; for an honest signature, τ is the restriction of the dual $\widehat{\sigma}$ of the response. Let $\lambda \equiv q^{-1} \pmod{2^r}$. The base points are R and S on copies 0 and 1, and $\tau(R)$ and $\lambda\tau(S)$ on copies 2 and 3: write b_n, for $0 \le n \le 7$, for the base point $\lfloor n/4 \rfloor$ on copy $n \bmod 4$, so that a coefficient vector $v \in (\mathbb{Z}/2^r)^8$ stands for the point $\sum_n v_n b_n$ of $A[2^r]$, and a matrix acting on coefficient vectors has as column n the image of b_n. Let $\Omega = \left(\begin{smallmatrix} 0 & I \\ -I & 0 \end{smallmatrix}\right)$, with 4×4 blocks, and $\omega(u, v) = u^{\mathsf{T}} \Omega v$.

Lemma 10 (Base points)**.** *Suppose that q is odd and that the rule of Section 4 is defined, that is, the entry it inverts is odd. Then $e(\tau R, \tau S) = e(R, S)^q$, the points $b_0, \dots, b_7$ form a basis of $A[2^r]$, and, for the pairing e_A of the product polarization and $\zeta = e(R, S)$,*

$$e_A\Big(\sum_n u_n b_n, \sum_n v_n b_n\Big) = \zeta^{\omega(u,v)}.$$

In particular, a matrix over $\mathbb{Z}/2^r$ is symplectic for ω exactly when its columns, applied to the base points, form a ζ-symplectic basis of $A[2^r]$.

Proof. The rule enforces $ad - bc \equiv kq \pmod{2^r}$, so by bilinearity $e(\tau R, \tau S) = e(P, Q)^{ad-bc} = e(P, Q)^{kq} = e(R, S)^q$. Hence $e(\tau R, \lambda\tau S) = \zeta^{q\lambda} = \zeta$, which has order 2^r, so τR and $\lambda\tau S$ form a basis of $E_{\mathrm{chl}}[2^r]$. The pairing e_A is the product of the pairings on the four copies, so $e_A(b_n, b_{n+4}) = \zeta$ for $0 \le n \le 3$, while two base points on different copies, or equal ones, pair to 1. Bilinearity gives the formula. □

When q is even, or the entry the rule inverts is even, the rule is not defined and the verifier rejects; such inputs are not among those for which the half-chains are defined.

The Kani matrices Since $q\lambda \equiv 1$, we have $\tau(xR + yS) = x \cdot \tau R + qy \cdot \lambda\tau S$. Let $\widehat{\tau}\colon E_{\mathrm{chl}}[2^r] \to E_{\mathrm{com}}[2^r]$ map $u \cdot \tau R + v \cdot \lambda\tau S$ to $qu \cdot R + v \cdot S$; then $\widehat{\tau}\tau = q$ and $\tau\widehat{\tau} = q$, and for an honest signature $\widehat{\tau}$ is the restriction of σ. With $T = \mathrm{diag}(\tau, \tau)$ and $\widehat{T} = \mathrm{diag}(\widehat{\tau}, \widehat{\tau})$, the verifier uses, on $A[2^r] = E_{\mathrm{com}}^2[2^r] \times E_{\mathrm{chl}}^2[2^r]$,

$$F = \begin{pmatrix} \alpha & \widehat{T} \\ -T & \widehat{\alpha} \end{pmatrix}, \qquad \widehat{F} = \begin{pmatrix} \widehat{\alpha} & -\widehat{T} \\ T & \alpha \end{pmatrix},$$

which for an honest signature are the restrictions to $A[2^r]$ of the endomorphisms F and $\widehat{F}$ of Section A.2 with the two curves exchanged, as stated there. The kernel lifts of Lemma 9 are the images under $\widehat{F}$ of b_0, b_4, b_1 and b_5, in that order.

Lemma 11 (The Kani matrices)**.** *On coefficient vectors, $F = \mathrm{diag}(F_1, F_2)$ and $\widehat{F} = \mathrm{diag}(\widehat{F}_1, \widehat{F}_2)$, where $F_1, \widehat{F}_1$ act on the coefficients of the first base points (rows 0 to 3) and $F_2, \widehat{F}_2$ on those of the second (rows 4 to 7), and, in 2×2 blocks with α, $\widehat{\alpha}$ the integer matrices of Lemma 1 and an integer standing for that multiple of I_2,*

$$F_1 = \begin{pmatrix} \alpha & q \\ -1 & \widehat{\alpha} \end{pmatrix}, \quad F_2 = \begin{pmatrix} \alpha & 1 \\ -q & \widehat{\alpha} \end{pmatrix}, \quad \widehat{F}_1 = \begin{pmatrix} \widehat{\alpha} & -q \\ 1 & \alpha \end{pmatrix}, \quad \widehat{F}_2 = \begin{pmatrix} \widehat{\alpha} & -1 \\ q & \alpha \end{pmatrix}.$$

Let $\tilde{D}$ be the action of D on coefficient vectors (Lemma 9). The following identities hold over $\mathbb{Z}[a_1, a_2, q]$, the first with $a_1^2 + a_2^2 = 2^e - q$ by the choice of a_1, a_2:

$$\begin{gathered} F\widehat{F} = \widehat{F}F = (a_1^2 + a_2^2 + q)\, I = 2^e I, \qquad \omega(Fu, v) = \omega(u, \widehat{F}v), \\ F\tilde{D} = \tilde{D}F, \qquad \widehat{F}\tilde{D} = \tilde{D}\widehat{F}, \qquad \omega(\tilde{D}u, \tilde{D}v) = \omega(u, v). \end{gathered} \tag{1}$$

Proof. On the first coefficients of each copy, τ acts as 1 and $\widehat{\tau}$ as q; on the second coefficients, τ acts as q and $\widehat{\tau}$ as 1. Substituting these into F and $\widehat{F}$ gives the blocks. Since $\alpha\widehat{\alpha} = \widehat{\alpha}\alpha = a_1^2 + a_2^2$ (Lemma 1(iii)) and integers commute with everything, multiplying the 2×2 blocks gives $F_i\widehat{F}_i = \widehat{F}_i F_i = (a_1^2 + a_2^2 + q)I$ for $i = 1, 2$. The blocks satisfy $\widehat{F}_2 = F_1^{\mathsf{T}}$ and $\widehat{F}_1 = F_2^{\mathsf{T}}$, because $\widehat{\alpha} = \alpha^{\mathsf{T}}$; this says $F^{\mathsf{T}}\Omega = \Omega\widehat{F}$, that is, $\omega(Fu, v) = \omega(u, \widehat{F}v)$. On coefficient vectors $\tilde{D} = \mathrm{diag}(K, K)$ with $K = \mathrm{diag}(J, J)$ and $J = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ (Lemma 9). The blocks $\alpha = a_1 + a_2 J$ and $\widehat{\alpha} = a_1 - a_2 J$ commute with J, and so do integers, so K commutes with F_1, F_2, $\widehat{F}_1$ and $\widehat{F}_2$. Finally $K^{\mathsf{T}}K = I$, so $\tilde{D}^{\mathsf{T}}\Omega\tilde{D} = \Omega$. □

In particular the first kernel block of Lemma 9 is $X^{(1)} = \widehat{F}E$, where E is the matrix with columns the standard basis vectors e_0, e_4, e_1 and e_5, in that order. We have $e \geq r$: $r = \lceil e/2 \rceil + 2 \leq e$ as soon as $e \geq 4$, and $(e, r) = (174, 89)$, $(264, 134)$ and $(346, 175)$ at levels I, III and V (Table 1). So the matrix $2^e I$ vanishes modulo 2^r.

A starting matrix M has as columns 0 to 3 the completion Y and as columns 4 to 7 the kernel lifts X; by Lemma 10 its columns, applied to the base points, give a symplectic basis of $A[2^r]$ exactly when M is symplectic for ω.

Lemma 12 (Any completion)**.** *Let $M = (Y \mid X)$ be symplectic over $\mathbb{Z}/2^r$, that is, $\omega(Y, X) = I$ and $\omega(Y, Y) = \omega(X, X) = 0$ column by column, and suppose that $\tilde{D}X = XU$ for a signed permutation matrix U. Then the columns of X span a Lagrangian direct summand of $(\mathbb{Z}/2^r)^8$, and $\tilde{D}Y = YU + XS$ with $S = \omega(Y, \tilde{D}Y - YU)$, a 4×4 matrix over $\mathbb{Z}/2^r$.*

Proof. A symplectic matrix is invertible, so its columns form a basis, and the span of X is a free direct summand with complement the span of Y. It is isotropic, since $\omega(X, X) = 0$. If $v = Y\mathbf{a} + X\mathbf{b}$ is orthogonal to every column of X, then $0 = \omega(v, X) = \mathbf{a}^{\mathsf{T}}$, so v lies in the span of X: the summand is its own orthogonal, that is, Lagrangian. Let $\Delta = \tilde{D}Y - YU$. Since $\tilde{D}$ preserves ω and $U^{\mathsf{T}}U = I$,

$$\omega(\Delta, XU) = \omega(\tilde{D}Y, \tilde{D}X) - U^{\mathsf{T}}\omega(Y, X)\,U = I - U^{\mathsf{T}}U = 0,$$

and U is invertible, so every column of Δ is orthogonal to the span of X and lies in it: $\Delta = XS$, and $\omega(Y, \Delta) = \omega(Y, X)S = S$. □

Lemma 13 (The generators are paired)**.** *For the two starting matrices built by the verifier, $M_1 = (Y^{(1)} \mid -\widehat{F}Y^{(2)})$ and $M_2 = (Y^{(2)} \mid X^{(2)})$, with any symplectic completions $Y^{(1)}$, $Y^{(2)}$, both matrices are symplectic, and $\tilde{D}$ acts on the kernel columns of each by the same signed permutation U, exactly modulo 2^r: it maps the first kernel column to minus the third, the second to minus the fourth, the third to the first and the fourth to the second. Consequently $Dg_0 = -g_2$, $Dg_1 = -g_3$ as points of order 2^r, for the generators of both half-chains, and $\tilde{D}Y^{(h)} = Y^{(h)}U + X_h S_h$ for $h = 1, 2$, where $X_1 = -\widehat{F}Y^{(2)}$, $X_2 = X^{(2)}$ and S_h is as in Lemma 12.*

Proof. The verifier builds its matrices in four steps; we follow them. (1) The first kernel block $X^{(1)}$ is that of Lemma 9, paired by U; its columns are columns of $\widehat{F}$, so $FX^{(1)} = F\widehat{F}E = 2^e E = 0$, with E as after Lemma 11. (2) Its completion $Y^{(1)}$ satisfies $\tilde{D}Y^{(1)} = Y^{(1)}U + X^{(1)}S_1$ by Lemma 12. (3) The second kernel block is $X^{(2)} = FY^{(1)}$, and $\tilde{D}X^{(2)} = F\tilde{D}Y^{(1)} = X^{(2)}U + FX^{(1)}S_1 = X^{(2)}U$. It is isotropic, since $\omega(FY^{(1)}, FY^{(1)}) = \omega(Y^{(1)}, \widehat{F}FY^{(1)}) = 0$, and its completion $Y^{(2)}$ satisfies $\tilde{D}Y^{(2)} = Y^{(2)}U + X^{(2)}S_2$, again by Lemma 12. (4) The first half-chain's kernel columns are $X_1 = -\widehat{F}Y^{(2)}$, and $\tilde{D}X_1 = -\widehat{F}Y^{(2)}U - \widehat{F}FY^{(1)}S_2 = X_1 U$, since $\widehat{F}F = 2^e = 0$. The matrix M_1 is symplectic: $\omega(Y^{(1)}, X_1) = -\omega(FY^{(1)}, Y^{(2)}) = -\omega(X^{(2)}, Y^{(2)}) = \omega(Y^{(2)}, X^{(2)})^{\mathsf{T}} = I$ and $\omega(X_1, X_1) = \omega(Y^{(2)}, F\widehat{F}Y^{(2)}) = 0$ by (1), while $\omega(Y^{(1)}, Y^{(1)}) = 0$ from step (2). Lemma 12 applied to M_1 and M_2 gives the last assertion. Applying the columns to the base points, which are the same on both copies of each curve, turns these identities into identities between points. □

Definition 2 (The algorithm's theta structures)**.** Let $e_1 = \lceil e/2 \rceil$ and $e_2 = e - e_1$ be the lengths of the two half-chains $\Phi_1 = F_1$ and $\Phi_2 = \widetilde{F}_2$ from A, with $F = F_2 \circ F_1$. For $h \in \{1, 2\}$, applying the columns of $2^{r-2-e_h} M_h$ to the base points gives, by Lemmas 10 and 13, a symplectic basis $(y'_{h,0}, \ldots, y'_{h,3}; x'_{h,0}, \ldots, x'_{h,3})$ of $A[2^{e_h+2}]$, with $\ker \Phi_h = \langle 4x'_{h,j} \rangle$. For $m \leq k \leq e_h$, with m as in Lemma 14 below, write $\Phi_{h,k}$ for the first k steps of Φ_h and

$A_{h,k}$ for its codomain, and put $x'_{h,k,j} = 2^{e_h-k}x'_{h,j}$, of order 2^{k+2}, and $y_{h,j} = 2^{e_h}y'_{h,j} \in A[4]$. The *algorithm's theta structure* on $A_{h,k}$ is the level-2 structure induced by

$$B_{h,k} = \big(\Phi_{h,k}(y_{h,0}), \dots, \Phi_{h,k}(y_{h,3});\ \Phi_{h,k}(x'_{h,k,0}), \dots, \Phi_{h,k}(x'_{h,k,3})\big).$$

These are the structures in which SQIsignHD's half-chain algorithm computes [**?**, Appendix F], [**?**, Section 5.3 and Appendix C], as an induction on k shows. Each half-chain starts with m steps in dimension 2, and the gluing step $m+1$ is computed in the structure induced by the symplectic basis $B_{h,m}$ of $A_{h,m}[4]$. This is [**?**, Appendix C.2.2 and Lemma 31] in our notation. There the symplectic basis of $A[2^r]$ given by M_h lists the completion first and the kernel lifts second, the kernel of Φ_h is 2^{r-e_h} times the span of the kernel lifts, and the gluing uses the basis formed by $2^{r-2}\Phi_{h,m}$ of the completion and $2^{r-2-m}\Phi_{h,m}$ of the kernel lifts, which is $B_{h,m}$. A step f from $A_{h,k}$ uses 8-torsion points T'' with $4T''$ spanning its kernel; completed to a symplectic basis $(S'';T'')$ of $A_{h,k}[8]$ above the basis inducing the structure of $A_{h,k}$, it gives the codomain the structure induced by the symplectic basis $(2f(S''); f(T''))$ [**?**, Sections 4.1 and 4.2]. This is [**?**, Theorem 56] in the form in which [**?**] states and applies it, with the half above the kernel second, as here. With $2S'' = \Phi_{h,k}(y_h)$ and $T'' = \Phi_{h,k}(x'_{h,k+1})$, that basis is $B_{h,k+1}$. Theorem 2 below is a statement about this algorithm.

Lemma 14 (The gluing codomain)**.** *For every input for which the verifier's half-chains are defined, a_1 is odd, a_2 is even and nonzero, and the dimension-4 gluing is step $m+1$ of each half-chain, with $m = v_2(a_2) \geq 1$. So the gluing codomain is $A_{h,m+1}$, with $m+1 \geq 2$.*

Proof. The algorithm takes q odd and names the odd one of a_1, a_2 as a_1, which is possible because $a_1^2 + a_2^2 = 2^e - q$ is odd [**?**, Section 5.1 and Algorithm 6]. It computes the first $m = v_2(a_2)$ steps of each half-chain in dimension 2 and step $m+1$ as the dimension-4 gluing [**?**, Lemmas 23 and 28]. This needs m finite, that is, $a_2 \neq 0$, and then $m \geq 1$ because a_2 is even. For a good degree nothing is excluded: $2^e - q$ is then a prime, which is not a square, so $a_2 \neq 0$. □

Theorem 2 (The base case)**.** *For every input for which the verifier's half-chains are defined, honest or not, with any symplectic completions, for $h \in \{1,2\}$ and every k with $\max(2,m) \leq k \leq e_h$, the automorphism $D_{h,k}$ of $A_{h,k}$ induced by D acts on the coordinates of the algorithm's theta structure on $A_{h,k}$ as the permutation π: there is $\mu \in \Bbbk^*$ with $\theta_a(D_{h,k}z) = \mu\,\theta_{\pi(a)}(z)$ for all $z \in A_{h,k}$ and all a. In particular the hypotheses of Theorem 1 hold for both half-chains, and from the gluing codomain on, the two-point algorithm computes the same kernels, isogenies and codomains as the four-point algorithm.*

Proof. Fix h and let U be the signed permutation of Lemma 13. The kernel of $\Phi_{h,k}$ is generated by the $4x'_{h,k,j}$ and is stable under D by Lemma 13, so the proof of Proposition 2 gives an automorphism $D_{h,k}$ with $D_{h,k} \circ \Phi_{h,k} = \Phi_{h,k} \circ D$, which preserves the principal polarization of $A_{h,k}$ (as Proposition 4 notes, that proof uses only the stability of the kernel).

Multiplying the identity $\tilde{D}X_h = X_hU$ of Lemma 13 by 2^{r-2-k} and applying it to the base points gives $Dx'_{h,k} = x'_{h,k}U$ exactly, hence $D_{h,k}\Phi_{h,k}(x'_{h,k}) = \Phi_{h,k}(x'_{h,k})\,U$. Multiplying $\tilde{D}Y^{(h)} = Y^{(h)}U + X_hS_h$ by 2^{r-2} in the same way gives $Dy_h = y_hU + x_hS_h$, where $x_{h,j} = 2^{e_h}x'_{h,j}$. The point $4x'_{h,k,j} = 2^{e_h-k+2}x'_{h,j}$ lies in the kernel of $\Phi_{h,k}$, and $x_{h,j}$ is a multiple of it as soon as $k \geq 2$; then $\Phi_{h,k}(x_{h,j}) = 0$ and $D_{h,k}\Phi_{h,k}(y_h) = \Phi_{h,k}(y_h)\,U$. So $D_{h,k}B_{h,k} = B_{h,k}\,\mathrm{diag}(U,U)$, exactly.

The basis $B_{h,k}$ is symplectic for a primitive fourth root of unity (paragraph after Definition 2), so Lemma 7 now gives $\theta_a(D_{h,k}z) = \mu\,\theta_{U^{-1}a}(z)$. Modulo 2, $U^{-1} = U^{\mathsf{T}}$ is the permutation matrix of (0 2)(1 3), since U maps the first kernel generator to minus the third and the second to minus the fourth, and the third and fourth to the first and second.

By the index convention of Section A.5, the index map $\mathbf{i} \mapsto U^{-1}\mathbf{i} \bmod 2$ exchanges bits 0 and 2 and bits 1 and 3: it is π.

For the last assertion, Lemma 13 gives the pairing of the generators, and Lemma 14 shows that the gluing codomain is $A_{h,m+1}$ with $m+1 \geq 2$, where the block action has just been proved; Theorem 1 applies. □

The bound $k \geq 2$ matters only when $m = 1$. Then on $A_{h,1}$ the term $x_h S_h$ need not vanish under $\Phi_{h,1}$: $D_{h,1}$ acts on $B_{h,1}$ by $\left(\begin{smallmatrix} U & 0 \\ 2S_h & U \end{smallmatrix}\right)$ modulo 4, and the first part of Lemma 7 with Lemma 6(iii) gives $\theta_{U\mathbf{i}}(D_{h,1}z) = \mu\,(-1)^{\langle U\mathbf{i}|S_h\mathbf{i}\rangle}\,\theta_{\mathbf{i}}(z)$, the permutation twisted by signs. This does not affect the verifier, whose two-point chain starts at the gluing codomain $A_{h,m+1}$.

Proposition 5 (The second half-chain)**.**

(i) The second half-chain is the chain of $\widetilde{F}_2$, computed from A by the same algorithm from the second starting matrix; its theta structures are those of Definition 2 with $h = 2$, which [?, Theorem 56] induces from that matrix, and Theorem 2 covers it. The verifier evaluates F_2 as the dual of this chain, step by step; the two-point algorithm changes nothing there, since it computes the same steps.

(ii) Suppose $F = F_2 \circ F_1$, as for honest signatures. The starting matrices satisfy the conditions of [?, Corollary 58], with $(x'; y') = (-x'_1; y'_1)$ on the side of F_1 and $(u'; v')$ given by M_2 on the side of $\widetilde{F}_2$, applied to the base points scaled to order 2^{e_1+2}. Hence, by [?, Proposition 57], the structures that the two half-chains induce on their common codomain C have bases related by the exchange of halves $\left(\begin{smallmatrix} 0 & -I \\ I & 0 \end{smallmatrix}\right)$, and their coordinates differ by a Hadamard transform [?, Lemma 29 and proof of Lemma 22]. The automorphisms of C induced by D along the two half-chains coincide, and π commutes with the Hadamard transform, so the two instances of Theorem 2 on C agree.

Proof. (i) The kernel lifts and the completion of M_2 play the roles that those of M_1 play for the first half-chain, and the proof of Theorem 2 used nothing else about them.

(ii) Write $d' = 2^{e_1}$, $c_1 = 1$ and $c_2 = 2^{e_1 - e_2}$, as in [?, Proposition 57]. The basis $(-x'_1; y'_1)$ is ζ-symplectic because $(y'_1; x'_1)$ is, and $\ker F_1 = \langle 4c_1(-x'_1)\rangle$; on the other side $\ker \widetilde{F}_2 = \langle 4x'_2\rangle = \langle 4c_2 v'\rangle$. The conditions of Corollary 58, $F(c_2 y') = c_2 v'$ and $\widetilde{F}(c_1 u') = c_1 x'$, follow from $X^{(2)} = FY^{(1)}$ and $-X_1 = \widehat{F}Y^{(2)}$ in Lemma 13, with $\widetilde{F} = \widehat{F}$ by (1). Proposition 57 orders the basis of F_1 with the half above the kernel first and that of $\widetilde{F}_2$ with it second; the algorithm uses the second order on both sides, which exchanges the two halves of the first basis and gives the matrix stated. If $F_2 \circ F_1 = -F$ instead, replacing F_2 by $-F_2$ changes neither kernel nor the structures, since a basis and its negative induce the same structure.

Let D_C satisfy $D_C F_1 = F_1 D$. Then $F_2 D_C F_1 = FD = DF = DF_2 F_1$, and F_1 is surjective, so $F_2 D_C = DF_2$. Taking duals, with $\widehat{D} = D^{-1}$ and $\widehat{D_C} = D_C^{-1}$ (Proposition 2), gives $D_C^{-1}\widetilde{F}_2 = \widetilde{F}_2 D^{-1}$, that is, $D_C \widetilde{F}_2 = \widetilde{F}_2 D$: D_C is also the automorphism induced along $\widetilde{F}_2$. Finally, π permutes index bits, so $\langle \pi(a)|\pi(b)\rangle = \langle a|b\rangle$ and π commutes with the Hadamard transform $\theta_a \mapsto \sum_b (-1)^{\langle a|b\rangle}\theta_b$. □

Remark 2 (An equivariant completion). One can also choose the completion so that $\tilde{D}Y = YU$ exactly: solve for two completion columns, adjust each by one kernel column so that $\omega(y, \tilde{D}y) = 0$, which is possible because $\omega(y, \tilde{D}y)$ is always even in these coordinates, and obtain the other two as $-\tilde{D}$ of the first two. Then $S = 0$ in Lemma 12. The implementation provides this completion and uses it in the two-point verifier; by Lemma 13 it is not needed for Theorem 2.

A.7 Conformance of the implementation

Theorem 2 is a statement about the verifier's algorithm. It applies to our verifier to the extent that the implementation computes that algorithm. This subsection separates what we established about that by reading the code of our crate from what remains unproven, which is set in red.

Remark 3 (Conformance). The following hold by inspection of our crate's code.

(a) *Index convention.* Our crate maps the multi-index (i_0, i_1, i_2, i_3) to $i_0 + 2i_1 + 4i_2 + 8i_3$, and its change of theta coordinates multiplies component l of the multi-index by column l of each block, with the blocks of [**?**, Theorem 12] in the positions $\left(\begin{smallmatrix} U & V \\ W & U' \end{smallmatrix}\right)$ of Lemma 6. It takes the kernel generators from columns 4 to 7 of the starting matrices, in that order, and places copies 0 and 1 on E_{com} and copies 2 and 3 on E_{chl}. The coordinate permutation of the two-point verifier puts $\theta_{\pi(a)}$ in position a. So bit l is labeled by g_l, as Section A.5 assumes, and the permutation used is π.

(b) *The index* $\mathbf{i}_0$. Our crate's change of theta coordinates takes for $\mathbf{i}_0$ the first multi-index, in the order $0, 1, \ldots, 15$, whose coefficient vector $T_{\mathbf{i}_0}$ is nonzero. It tests the condition on sections, as Lemma 6 requires, and takes $\mathbf{i}_0 = 0$ whenever $V = 0$.

(c) *The integers* a_1, a_2. Our crate's Cornacchia step returns the odd value as a_1, and rejects, among other cases, when $2^e - q$ is even or not congruent to 1 modulo 4. It computes m as the number of trailing zero bits of a_2, which is 128 when $a_2 = 0$, and rejects every $m > 40$. So every input that reaches our half-chains has a_1 odd and $1 \le m \le 40$, as Lemma 14 requires. Before Cornacchia's algorithm, our verifier repeats the goodness test of Section 4 and rejects a degree that is not good; being deterministic, it runs its 40 Miller–Rabin rounds on the first 40 primes as bases. The properties above are all that this appendix uses.

(d) *Projective kernel points.* Our crate computes the codomain of a step, and the images under it, from each kernel point only through ratios of that point's own coordinates, so a kernel point may be replaced by any nonzero multiple of its coordinate vector, as the proof of Theorem 1 assumes.

(e) *Base points and the dropped entry.* Our crate reads the transmitted matrix as the map τ of Section A.6, from the commitment basis to points of E_{chl}, uses the base points of Lemma 10 with $\lambda \equiv q^{-1}$, and recovers the dropped entry by the rule of Section 4, rejecting a signature whose transmitted a and b are both even, where the rule is not defined. Every input that reaches our half-chains therefore satisfies the hypotheses of Lemma 10.

(f) *Our Rust port.* Of our Rust port we established the same items by reading its code at the commit of the session of Section 5: the index convention of (a) line by line against the appendix, the index $\mathbf{i}_0$ of (b), the projective use of the kernel points of (d), the base points and the rule of (e), and, for (c), the odd a_1, the parity and residue tests on $2^e - q$, and the bound on m. The port's verifier does not repeat the goodness test before Cornacchia's algorithm: the only tests on $2^e - q$ are the parity, the residue modulo 4 and the success of the descent. It rejects a transmitted a and b both even before any inversion. The dimension-2 gluing is one chain applied to both pairs of copies; whether its structure, the dimension-4 gluing's change of basis and the plain steps are those of Definition 2 is not established for the port either, and what the port's tests show is only that its two-point and four-point chains agree at every step and that the SQIsignHD library's verifier accepts its signatures.

What is not established is that our crate computes the algorithm's theta structures of Definition 2 on every fourfold $A_{h,k}$ with $k \ge m$: that is, that its dimension-2 gluing

$\Phi\colon E_{\mathrm{com}} \times E_{\mathrm{chl}} \to A_m$, applied with the same theta structure to both pairs of copies, its change of basis for the dimension-4 gluing, its dimension-4 gluing step and the plain steps after it are faithful. Our crate computes the change of basis of the gluing by closed formulas taken from [**?**] (cf. [**?**, Lemmas 30 and 31]), meant to express the basis $B_{h,m}$ in the product theta basis of $A_m \times A_m$; we have not verified them against Definition 2. We checked only that on 60 inputs, from 24 honest and 36 forged signatures, our crate returns exactly the matrices that the formulas of [**?**] return on the same inputs, all of them symplectic modulo 4, with m ranging over 1, 2, 3, 4, 6 and 8. The dimension-2 gluing, the dimension-4 gluing step and the plain steps are not checked. Until they are, Theorem 2 reaches our verifier only through this statement.

Evidence Everything observed agrees with Theorem 2. Solving for the matrix N_k with $N_k\,\theta(x) \sim \theta(Dx)$ from 20 random points at the gluing codomain and after plain steps 1, 2, 3, 20, 43, 70 and 84 of the first half-chain of one level-I signature gives the pure permutation π, with all scalars equal to 1, and nothing else. The identity $\theta(Dx) = \pi(\theta(x))$ held at every step of both half-chains, on four random points per signature and on every codomain null point: with the generic completion, for 32 honest signatures (5 472 steps) and the 72 of 96 forged ones that pass the setup (12 318 steps); with the equivariant one, for the same honest signatures and 70 forged ones from other seeds (17 332 steps in all). On these 102 signatures, $\tilde{D}$ acts on the generic starting matrices exactly as Lemmas 12 and 13 say, modulo 4: block-diagonally on the kernel lifts and modulo the kernel on the completion, although not block-diagonally on the completion itself in 47 of them; and the gluing's change of basis intertwines D, acting on $A_m \times A_m$ as the signed exchange of the two copies of A_m, which is how J acts on a square, with a block-diagonal signed permutation, while an unsigned exchange fails on all 102. Levels III and V, which are not implemented, remain untested.

What the implementation does meanwhile The two-point verifier checks on each signature that the gluing images of g_2, g_3 are the π-permutations of those of g_0, g_1, and falls back to four points otherwise; the fallback never occurred on the signatures tested. Until the conformance of Remark 3 is established this check should stay; it costs four evaluations through the gluing.

A.8 Beyond two-point evaluation

The two-point verifier halves the work of doubling and pushing kernel points; it still computes each codomain from four points. MIKE also computes the codomain from two points [**?**, Theorem E.1], using coordinates adapted to its automorphism in the sense of [**?**, Definition G.1] and a nonvanishing assumption on some theta coordinates [**?**, Heuristic E.1], and gives doubling formulas on the quotient by it. Transferring those would require carrying adapted coordinates through the gluing steps from a product of four elliptic curves, which differ from MIKE's, and checking the nonvanishing assumption on this family of fourfolds; a forged signature could be built to make one of those coordinates vanish, so a verifier using them must reject rather than divide by zero. None of this is done here.